Lab Report
CSE 302 – Computer Organization & Architecture

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Task#1: Write a program that asks the user to enter an integer and then displays the number of 1's in the binary representation of that integer. For example, if the user enters **9**, then the program should display **2**.

```
.data
msg1: .asciiz "Enter number: "
msg2: .asciiz "Number of 1 bits: "
.text
main:
li $v0,4
la $a0,msg1
syscall
li $v0,5
syscall
move $t0,$v0
li $t1,0
li $t2,0
loop:
beq $t2,32,done
andi $t3,$t0,1
add $t1,$t1,$t3
srl $t0,$t0,1
addi $t2,$t2,1
j loop
done:
li $v0,4
```

```
la $a0,msg2
syscall
li $v0,1
move $a0,$t1
syscall
li $v0,10
syscall
```

Task#2: Write a program that asks the user to enter two integers: **n1** and **n2** and prints the sum of all numbers from **n1** to **n2**. For example, if the user enters **n1=3** and **n2=7**, then the program should display the sum as **25**.

```
.data
m1: .asciiz "Enter n1: "
m2: .asciiz "Enter n2: "
m3: .asciiz "Sum = "
.text
main:
li $v0,4
la $a0,m1
syscall
li $v0,5
syscall
move $t0,$v0
li $v0,4
la $a0,m2
syscall
li $v0,5
syscall
move $t1,$v0
```

```
li $t2,0
loop2:
bgt $t0,$t1,done2
add $t2,$t2,$t0
addi $t0,$t0,1
j loop2
done2:
li $v0,4
la $a0,m3
syscall
li $v0,1
move $a0,$t2
syscall
li $v0,10
syscall
```

Task#3: Write a program that asks the user to enter an integer and then display the hexadecimal representation of that integer.

```
.data
m1: .asciiz "Enter number: "
m2: .asciiz "Hex = 0x"
.text
main:
li $v0,4
la $a0,m1
syscall
li $v0,5
syscall
```

```
move $t0,$v0


li $v0,4
la $a0,m2
syscall
li $v0,34
move $a0,$t0
syscall
li $v0,10
syscall
```